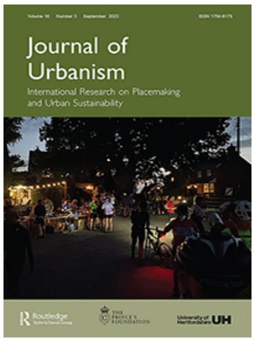

Journal of Urbanism: International Research on Placemaking and Urban Sustainability

ISSN: 1754-9175 (Print) 1754-9183 (Online) Journal homepage: www.tandfonline.com/journals/rjou20

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JOURNAL OF URBANISM
2023, VOL. 16, NO. 3, 310–340
https://doi.org/10.1080/17549175.2021.1987300

Women's mobilities and perceived safety: urban form matters. Evidence from three peripheral districts in the city of Bogotá

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ABSTRACT

The paper aims at investigating the interplay of urban form and women's mobilities in three peripheral districts in the city of Bogotá. Integrating a morpho-functional analysis of the built environment with an ethnographic analysis focused on the walking practices of a sample of women, the paper highlights the main gendered spatial experiences and how the perceived safety acts as a mediator between built environment attributes and walking behaviours. The conclusion introduces two challenges: first, the need for new interpretative lenses, different from those of a man-centred perspective and able to interpret the women's *territoriality* – considered as the spaces produced through their mobility practices; second, the possibility to use women's mobilities experiences and their tactics of adaptation as a tool for more effective urban and mobility policies.

KEYWORDS

Women's mobilities; gendered mobilities; morpho-functional patterns; safety perceptions; built environment; walkability; ethnographic analysis

Introduction

Gender is one of the main factors that influence individuals' travel behaviours and mobility practices. The rich body of work on mobility and gender deals with different forms of this influence, attempting to shed light on how gender – together with other socioeconomic, cultural and demographic features – defines diverse forms of mobility and should, therefore, be considered when planning transport (Marolda 2019; Law 2016; Scholten and Joelsson 2019; Uteng, Christensen, and Levin 2020; Uteng and Cresswell 2008; Walsh 2002). Even leaving aside the relevant issue of how mobility shapes gender and focusing exclusively on how gender shapes mobility (Hanson 2010), significant "gendered differences" (Uteng and Cresswell 2008, 2) emerge that have been investigated mainly in relation to women. These differences, already visible in childhood (Brown et al. 2008), are amplified in adulthood, when women often deal with a dual role as wage earners and domestic workers, frequently in charge of care-related activities. Determining specific mobility possibilities and constraints, often characterised by specific forms of

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deprivation (Dobbs 2007; Pickup 1984), these mobility patterns result more challenging to represent with traditional analytical approaches. In fact, while men tend to commute from home to their workplace and back, women develop varied forms of trip chaining in order to accomplish care-related tasks (Gilow 2020; McGuckin and Murakami 1999; Sánchez de Madariaga and Zucchini 2019). Alongside the socio-demographic and family conditions, attitudes, skills, and supply, the built environment plays a crucial role in determining women's practices and experiences. It requires mixed methods and approaches capable of integrating socio-ethnographic dimensions with the qualities and features of the urban settlements and transport supply (Montoya-Robledo and Escovar-Álvarez 2020).

In the body of research exploring the interplay between the built environment and travel behaviour (Ewing and Cervero 2001; Handy, Cao, and Mokhtarian 2005), urban and transport studies have devoted less attention to the effects of the built environment on women's mobilities. The built environment is relevant in relation to both the experiences it determines and the constraints it puts to the achievement of gender role-related tasks, differently influencing men's and women's practices and activities in the urban space (Rose 1993; Townsend 1991). As for the first condition, the perception of safety is an important variable, determined not only by the design of a space but rather – and more importantly – by the fact that women judge their safety depending on the social relations that occur in public space and the dread that these may generate (Valentine 1990). As for the second element, design features determine different neighbourhood configurations that can enhance or impede the completion of specific tasks (Goddard et al. 2006). The built environment has thus a twofold role concerning women's mobilities. On the one hand, it considerably contributes to ensuring comfort and safety for women on the move, often becoming responsible for specific inequality forms (Figueroa Martínez and Waintrub 2015). On the other hand, it also strongly influences how women address the family, health and safety concerns related to their gender role (Handy 2006). Addressing the built environment becomes thus crucial for planning urban mobility in a gender-sensitive way.

The interplay of the built environment and women's mobilities needs to be considered in place. Different settings show both physical features and cultural characteristics that influence how women perceive the built environment and, consequently, decide how to move (Arellana et al. 2019); crucial in this sense is the role of walkability, that is, the features of the built environment that facilitate the presence of people walking or spending time in the street space (Arellana et al. 2021). In general, mobility as a gendered phenomenon has been investigated mainly in European and North American settings, even if countries of the Global South show specific social, spatial and cultural features related to this issue (Priya Uteng and Turner 2019). In doing so, analytical tools developed elsewhere do not adapt to include the local social, cultural, and infrastructural peculiarities that determine different travel needs (Steiniger, Rojas, and Vecchio 2019). Simultaneously, these tools do not consider that people with different ethnic backgrounds perceive differently the same walking environment (Adkins et al. 2019).

Even focusing solely on the built environment, and considering the relevance of the analytical concept of walkability in Latin American settings, where walking is often the primary modal choice, urban forms and public policies do not pay attention and always support this mode (Herrmann-Lunecke, Mora, and Sagaris 2020). Drawing on this, the paper proposes a context-sensitive empirical analysis of the relationships between morphological settlements and women's mobilities practices. Providing a gendered Latin

American perspective on the relationship between built environment and travel behaviour, the research focuses on the daily walking patterns of a group of female residents in three selected neighbourhoods in the city of Bogotá (Colombia), each of them characterised by peculiar morphological and functional patterns. Developing a series of diaries based on the women's perceptions and experiences, the paper subsequently compares the interviewees' information with the built environment configuration of each neighbourhood. Integrating a morpho-functional analysis of the built environment with an ethnographic analysis conducted through travel along and interviews with a sample of women, the paper examines the built environment features that condition women's mobilities, highlighting specific impeding conditions of insecurity. These elements can provide useful elements for addressing gender-sensitive urban and mobility policies.

The built environment, walkability and travel behaviour: a gendered Latin American perspective

Research on gendered mobilities in Latin America does not give much explicit space to the role of the built environment and walkability in particular. Research on the relationship between women and mobility has focused on three main streams of research, referring to safety, violence and mobility, emphasising harassment and abuse; time use and interdependency; differentiated use of public space and transport supply, especially concerning the use of public transport (Jirón and Zunino Singh 2017). This focus also leads to a prevalence of qualitative approaches to the topic, mainly rooted in the new mobilities paradigm (Sheller and Urry 2006, 2016). Nonetheless, walkability is a recurrent issue in the works dealing with women's mobility, in relation to four elements: accessibility, safety, comfort, and interdependency.

Accessibility, considered as the possibility of reaching key places and opportunities, involves walking and implies significant gender differences that have received only marginal attention. For example, quantitative research devoted to accessibility evaluations of transport systems in Latin American settings does not provide works on women and does not consider much the built environment's role (Vecchio, Tiznado-Aitken, and Hurtubia 2020). A gendered perspective on accessibility would be relevant, considering the importance of walking for the mobility of women: they realise a high number of walking trips, engaging in more, shorter trips in comparison to men (Sagaris and Tiznado-Aitken 2020); moreover, walking also becomes a relevant intermodal alternative when the waiting times of public transport become too long (Fleischer and Marín 2019). When focusing on walking as a modal alternative, a contradictory situation may emerge, as in the Colombian cities of Barranquilla and Soledad (Arellana et al. 2021): the areas with the higher walkability are low-income zones where fewer opportunities are available, while wealthier areas close to urban opportunities show built environment conditions that are less favourable for walking. Moreover, the built environment also affects the access to transit stops by walking and the consequent possibility to use other transport modes, with significant interpersonal differences based on gender, age and location of the examined population (Tiznado-Aitken, Muñoz, and Hurtubia 2018; Tiznado-Aitken et al. 2020). The low quality of the urban environment can affect travel comfort and safety, and both are features determining significant accessibility barriers.

The low quality of public space can lead to perceived reduced safety and consequently fear when using the street space. Such perception seems to affect the whole experience of women's mobility since it involves both the spaces of public transport (be them vehicles, stations or stops) and the proximity space at the neighbourhood scale, but ends up conditioning also the experience of those who move by bicycle (Díaz Vázquez 2017; Fleischer and Marín 2019; Soto Villagrán 2017). In the Latin American setting, safety – intended as the perceived risk for crime and not as its actual occurrence – appears to influence walkability more than sidewalk condition or attractiveness, two features that instead are crucial in developed countries (Arellana et al., 2020). While significant differences emerge between day-time and night-time walking (Loukaitou-Sideris 2006), in the reviewed works walking is associated with unsafety in both moments of the day. In line with this, policemen's presence and an increase in their number can increase walkability scores, especially in low-income areas (Larrañaga et al. 2019).

Similarly, the built environment also affects the comfort of the walking experience, considering the condition of the sidewalks and the topography of a place. The former can be a significant barrier to walkability, although improving sidewalks' quality can significantly enhance walkability (Larrañaga et al. 2019). Topography instead is more difficult to address but becomes more critical when fewer mobility alternatives are available. For example, this is the case for car-dependent high-income areas such as those of Bogotá, located in mountainous areas of the city: when low-income domestic workers need to reach them, the absence of public transport services forces them to walk, making walking trips more difficult and configuring thus a significant barrier for their mobility (Fleischer and Marín 2019).

Finally, interdependence emerges as a feature that explains the relevance of walking and potentially contributes to walkability. Women are often in charge of care-related tasks: the interdependency between different family members often obliges them to accomplish many tasks at the neighbourhood scale, but the features of public space and different spatial configurations condition the possibility to move for women and the people they accompany – be them children, elderlies or sick people – hindering the completion of care-related activities (Chaves et al. 2017; Figueroa Martínez and Waintrub 2015; Gutiérrez and Reyes 2017; Jirón and Gómez 2018). Moreover, the perception of unsafety in public space may also generate specific practices, as in women who arrange to walk together (Figueroa Martínez and Forray 2015). In synthesis, while specific public space features emerge only as one of the many elements that influence women's mobility practices, the relevant impacts of different built environment typologies suggest focusing on different physical configurations to understand gender and mobility in place better.

Bogotá as a relevant test-field

To address these concerns we select the city of Bogotá as the field of investigation since many of Bogotá's features are paradigmatic of the Latin American city. Bogotá is the main urban agglomeration of Colombia, and in the last decades, it experienced rapid, inconsistent urban growth. Due to an intense internal migratory dynamic, originated by both the transition out of an agricultural economy and the displacement of rural inhabitants who fled civil war, a massive number of new inhabitants settled in the city, resulting in a fast urban expansion that soon outpaced the planning forecasts (Salcedo Fidalgo 2015;

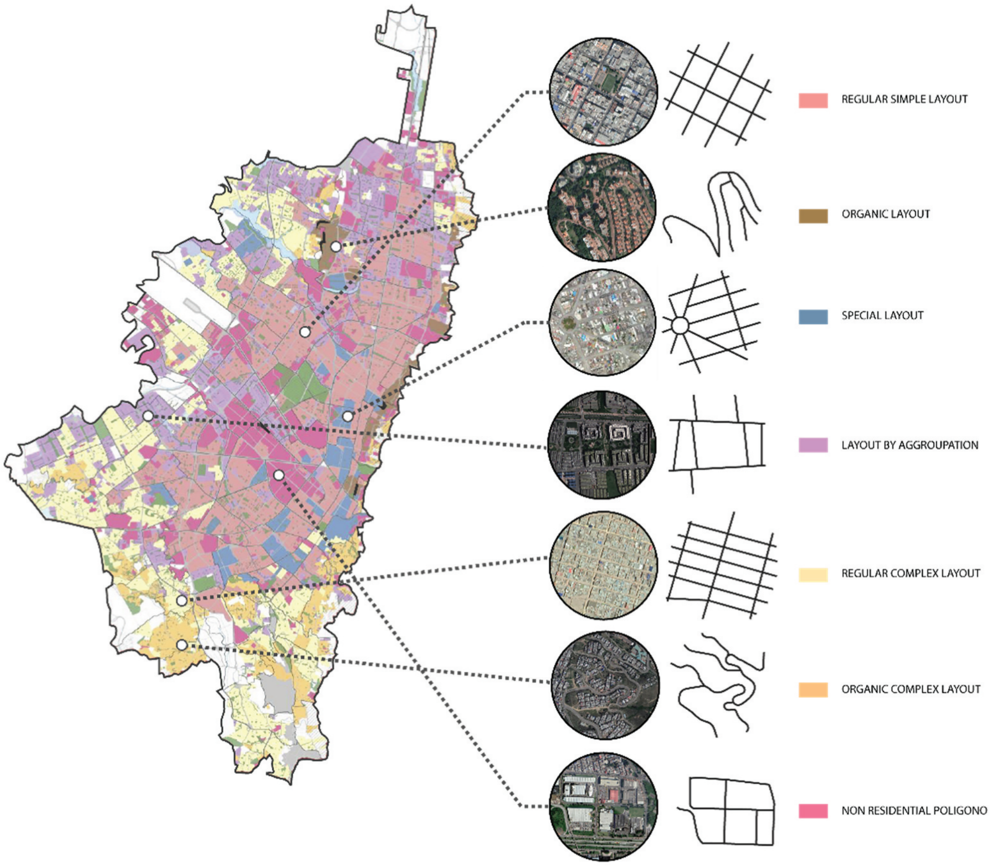

Figure 1. Spatial distribution of the morphological categories defined in the Land Use Plan (POT) of Bogotá (source: Plan de ordenamiento territorial. Documento técnico de soporte, 2018).

Torres Tovar 2009). The scattered growth of Bogotá resulted in an urban patchwork of different fragments, characterised by peculiar built environment features[1] (Figure 1) that significantly affect the quality of life of their inhabitants.

The effects of unplanned and scattered urban growth are especially visible in the city's peripheral areas, which grew informally and ended up hosting mainly low-income groups, reinforcing segregation dynamics (Gutiérrez 2011). These segregation dynamics have been present in the city since colonial periods, but they were reinforced by some zoning ordinances applied during the XXth century. During the 1930s the indigenous population, historically located on the eastern hills of the city, was forced to relocate to the south of the city with the creation of the first working-class neighbourhood (Saldarriaga, 2000). Simultaneously, upper-class families moved progressively to the north, where urban form conditions were based on the *City Beautiful* model. On the other hand, some land-use plans such as Plan Soto AaVv (2008), which established the west of the city as an industrial area and the south as a residential working-class area, introduced different built environment standards to those found in the north of the city.

The above, together with the housing demand produced by the increasing migration and the inefficient solutions provided by the private and public sector, generated a proliferation of illegal urbanizations, in which private individuals benefitted from the sale of illegal lots, notably concentrated in the south of the city. These neighbourhoods of illegal or semilegal origin were able to provide housing at a low cost, but the unplanned road network resulting from these informal developments affects both the mobility of the local inhabitants and the quality of the public spaces, and the street space in particular (Torres Tovar 2009). As a consequence, the periphery of the city nowadays is characterised by sub-standard housing, with population densities (i.e. number of people living in a unit area of land) higher than in the rest of the city, and by the inadequate provision of urban opportunities, infrastructures and transport supply (Bocarejo and Oviedo 2012; Guzman and Bocarejo 2017). Because of this, everyday mobility is a necessary and yet tricky practice, which devoted infrastructural interventions have only partially improved (Guzman and Oviedo 2018; Oviedo and Titheridge 2016; Vecchio 2017, 2020).

The morphological features and the spatial organisation of Bogotá make everyday mobility a difficult practice for most low-income inhabitants but affect especially women (Fleischer and Marín 2019). The role of women determines travel patterns and mobility practices that significantly diverge from the masculine ones. In Bogotá, 53,77% of the everyday trips are made by women, who also account for 60% of the total pedestrian trips in the city (Alcaldía Mayor de Bogotá 2017). Half of the women's trips are on foot, and a significant share of them lasts more than 15 minutes. These numbers result from the dual role of women as wage earners and domestic workers, often in charge of care-related activities, determining specific mobility possibilities and constraints, often characterised by specific forms of deprivation (Figueroa Martínez and Waintrub 2015).

This role leads to mobility patterns that differ from those of their male counterparts, and that are often more difficult to represent with traditional analytical approaches: while men tend to commute from home to their workplace and back, women develop varied forms of trip chaining in order to accomplish care-related tasks (Gilow 2020; McGuckin and Murakami 1999; Sánchez de Madariaga and Zucchini 2019). As a result, women in Bogotá dedicate on average more than 7 hours to household activities, while men barely dedicate an average of less than 3 hours and a half. In particular, if women travel less frequently than men, their trips take longer despite covering shorter distances (Lecompte & Bocarejo, 2017); low-income women account for the largest percentage of public transportation users (Montoya-Robledo et al., 2020) and in the case of domestic workers who travel in several neighbourhoods for their work, the lack of integrated fares within transportation system impacts their mobility capability (Oviedo and Titheridge 2016). In addition, women tend to have a lower purchasing power than men, making it difficult for them to use public transport and therefore access urban opportunities. The prevalence of care-related trips that occur by walking makes relevant to consider what are the features of the built environment through which women need to move and, consequently, to what extent these morphological features generate spatio-temporal barriers for their mobility.

Research design and study setting

In order to investigate the interplay of the built environment and travel behaviour, considering how it affects women's mobilities, we integrate a morpho-functional analysis of the built environment and an ethnographic analysis conducted through travel along and interviews with a sample of women living in three peripheral neighbourhoods of the city of Bogotá. We selected three neighbourhoods as case studies (Figure 2), according to two criteria:

- *socioeconomic conditions*. Using the official socioeconomic stratification for Colombia, which divides the population into six socioeconomic strata (being 1 the lowest and 6 the highest), areas belonging to the lowest socioeconomic levels were chosen. Low-income people tend to experience more significant difficulties in accessing the means of transport, also because they have less access to the car and are also less able to organise their daily journeys. Moreover, they are more exposed to crime, as reflected by the fact that affluent groups are willing to pay more for houses located in safer areas (Gaviria et al. 2010);

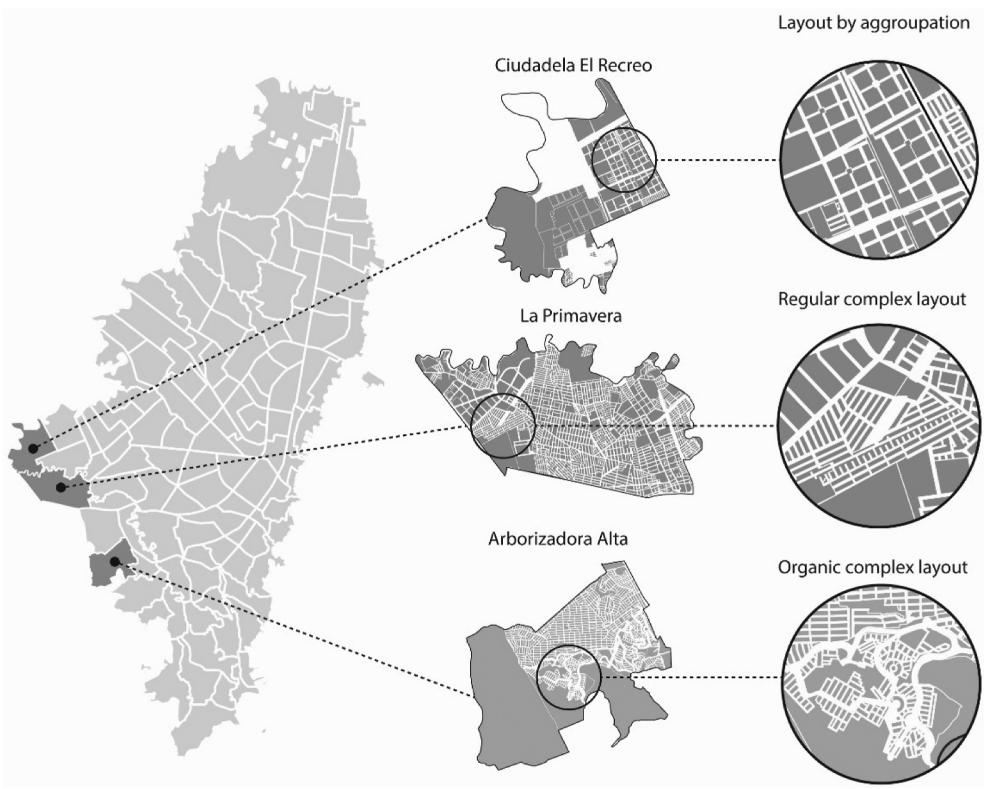

Figure 2. Selected neighbourhoods: location and morphological features (our elaboration).

Table 1. Selected neighbourhoods: morphological features and figures (source: Encuesta multipropósito 2017; Guzman and Bocarejo, 2017).

	Morphological type	Neighbourhood	Inhabitants		Population density (inhab./km2)	
			Total	% of women	Min	Max
Planned city	Layout by aggrupation (TPA)	Ciudadela El Recreo (UPZ Tintal Sur)	76.997	50,4%	15.000	25.000
Non-planned city	Complex regular layout (TRC)	La Primavera (UPZ Bosa Central)	286.334	50,4%	30.000	56.000
	Complex organic layout (TOC)	Arborizadora Alta (UPZ Jerusalén)	112.706	50,4%	25.000	56.000
City of Bogotá			7.181.569	52,2%	24,643	

- *urban morphology*. Based on the morphological units defined in the municipal land use plan of Bogotá (Table 1 and Figure 1), a sample of three different morphological configurations found on the South of the city was selected, belonging to both the formal and the informal city (Figure 2).

We addressed the selected case studies with a twofold method, based on a morphological analysis and on walking interviews.

First, we developed a morphological analysis of the three areas, referring to the "3Ds of the built environment" that convey the three dimensions through which the built environment influences travel demand (Cervero and Kockelman 1997). The approach is relevant for our analysis since it allows to consider the built environment in relation to its effects on how people decide to move. In particular, it considers the role of urban morphology[2] as the grounded outcome of social and economic forces, tangible shaping of intentions, and concrete transformations of the city (Moudon 1997), also in the case of informal settlements (Duarte 2009; Kamalipour 2016). The aim of the morphological analysis was to characterise each area according to their *density* (population density and built-up density), *diversity* (land use mix) and *design* (streets pattern, block size, permeability and pedestrian amenities like sidewalk quality, lighting, parks). The approach has envisaged a direct in-field survey in the selected areas, integrated by an urban morphological analysis (studying the urban fabric densities and their morphology, the open spaces and building relations, and functions for determining the human activity) through the available maps and data sources,[3] including Google Maps. The reconstruction of the neighbourhood planning processes provided further elements to complete this analysis.

Second, we integrated the morphological analysis with a series of extensive semi-structured walking interviews, submitted to a sample of female residents of each neighbourhood, along with a total of 12 diaries highlighting observations, perceptions and the reconstruction of daily life displacements. In each neighbourhood, we contacted four women through community action boards, selected based on age, family and working conditions. The sample for this investigation includes working women with children, both single and married, upon which their families rely for economic stability and fulfilment of household needs. The age of the interviewees varied between 18 years old to 50 years old.

The interviews were realised thank to a travel along approach: we accompanied each woman individually in their daily journeys and we collected the experiences thanks to a commented route. In doing so, each woman has been able to explain those aspects that she wants to highlight, in relation to building environment perception and walking patterns. The aim of the travel along method has been a repertoire of "perception in motion" because "three activities are solicited simultaneously: walking, perceiving and describing" (Thibaud 2001, 80). The collected experiences and perceptions of the built environment, conditioning the women's everyday mobility, were compared with the analysis conducted on the morphological characteristics of each neighbourhood, mapping the routes followed by the interviewees and the elements they mentioned, as well as developing a content analysis of their interviews looking for the elements referred to density, diversity and design of the three examined areas. Due to our interest in women's experiencies and perceptions, safety was a recurrent element in the interviews and received special attention in our analysis.

Approaching urban forms from a gender perspective

We analysed the morphological patterns of the three selected peripheral areas according to their *density* (population density and built-up density), *diversity* (land use mix) and *design* (streets pattern, block size, permeability and pedestrian amenities like sidewalk quality, lighting, parks) (Cervero and Kockelman 1997) as a framework for studying women's mobility habits in the informal communities in Bogota. Even if the literature explains well the relevance of the morphological patterns in influencing urban experiences, the role of density, design and diversity on the use of public spaces only marginally considers gender. A predominant perspective evaluates the impact of the built environment on women's mobilities as neutral, disregarding the role of gender perception, the importance of sensorial experiences and their implications on mobilities.

Starting with a selection of these studies, we try to understand the possible differential effects of the built environment on mobility practices, searching for analytical outcomes able to address the selection of relevant conditions affecting women experiences in the use of the city, to be tested in the field of Bogotá. In doing so, while several research papers highlight density, diversity and design as a crucial element of the urban form in order to address safety issues, their combined role is not always shared. According to Jacobs (1961), the continuous presence of people produced by the mix of primary uses (diversity and density) generates forms of natural surveillance, which increase a safety perception among citizens. In contrast with Jacobs´ ideas, other scholars have indicated that crime is more likely to occur in higher–density, mixed-use areas (Cohen and Felson 1979). These characteristics paradoxically can be associated with walkable environments, significantly correlated with the spatial layout of street networks. This aspect plays a vital role in establishing people's utilisation of places, movements and their perception of the surrounding environment, as well as the safety of a place (Johnson and Bowers 2010). Among the main features of the street networks, permeability has several implications on wayfinding opportunities and safety perceptions (Jacobs 1961; Newman 1996). The role of permeability can be relevant both for ordinary activities and emergencies: as for the former, a more permeable urban fabric gives the possibility to increase the existence of commerce spots (places for buying, eating, seeing things, getting a drink), increasing the flux of diverse people on the streets and in consequence, providing "eyes on the street" and catalysing diversity (Jacobs 1961; Newman 1996); as for the latter, permeability can enhance the three safety cues that streets can provide: prospect, refuge, and escape (Fisher and Nasar 1992).

However, there are few studies on the urban form tackling the safety problem from a gender perspective, despite its help to identify potentials and barriers, addressing alternative configurations of the public spaces and mobility supplies. Based on this, the proposed morphological analysis represents and describes the background on which subsequently we consider women's mobility experiences and habits. The organisation of the open spaces and services, the density of functions and persons offer the analytical framework for studying, through an ethnographic approach (travel along, diaries and interviews), how the women interviewed perceive the urban environment in which they live and travel.

Density, diversity and design in the three selected neighbourhoods

The morphological analysis of the urban settlement in the three selected areas in Bogotá, conducted through the available cartography and in fieldwork, allowed to define the main feature in term of densities of build-up areas, diversity in the land use and design of the open spaces and their equipment.

As for density, the three areas tend to show differences in built-up density, despite prevailing low-rise residential development and their population density is higher than the average of Bogotá (see Table 1). The neighbourhood la Ciudadela el Recreo, located in the Bosa District, was planned by the city administration to provide social housing to poor families and to mitigate the disadvantages of informal urbanisation. The project of this neighbourhood – designed by the temporal union of Gustavo Perry Arquitectos, Konrad Brunner Arquitectos y Eduardo Samper Martinez – was intended "*to enhance the concept of the grid, interweaving two layers, the vehicular and the pedestrian, on a principle of densification and qualification of the developable space represented in housing of high density and quality*" (Metrovivienda, 2011). The built-up density highlights heterogeneous situations inside this neighbourhood with 6-storey buildings along the main streets, and 2–3 storey buildings in the local streets characterised by a large section and several pedestrian paths. On the contrary, Arborizadora Alta in the Upz Jerusalem district is characterised by one of a lowest built-up density of the city (between 400.001 to 600,000 m2 per km2), along with a low rise residential development (25,001 to 55,900 inh/ km2). La Primavera is located in a high-density neighbourhood – the Upz Tintal Sur district – designed to provide housing to approx. 50.000 inhabitants and 240 single-family houses per block.

As for diversity, the three areas tend to show a high land use mix, although with some differences. Ciudadela El Recreo was intendedly designed as a residential community served by different facilities; in the neighbourhood, both a shopping centre and spontaneous family commercial activities (usually located on the ground floor of single-family houses) are present. La Primavera is characterised by a strong mix of residential and commercial land use, with a wide variety of commercial activities – such as convenience stores, retail services, supermarkets and restaurants; similarly, various schools and child

Figure 3. Diversity: land uses in the three selected neighbourhoods (our elaboration of data collected by the authors).

care facilities are locally available. Arborizadora Alta too is a mainly residential area, where various commercial facilities are present but only in few parts of the neighbourhood (Figure 3).

The design appears to be an element of differentiation among the three areas, due to their morphology and topographic conditions (Figure 4). In Ciudadela El Recreo, the street layout shows a continuous regular grid, composed of blocks with an approx area of 10.000 m^2 each-one, occupied by multi-family and single-family houses where low-income families prevail. The area has a system of local vehicular roads with unusually wide and well-maintained sidewalks. At the same time, different green areas are present in the neighbourhood, such as a linear park and smaller green areas located at the intersections of the internal pedestrian streets of each super-block. In La Primavera, the street network is regular and defines a grid composed of small blocks, each with a high occupancy rate (> 90% of the covered area). The quality of the public space is instead low since many streets are unpaved and some streets lack adequate sidewalks – most of them are too narrow or in some parts even missing. Finally, in Arborizadora Alta, the local topography determines a curvilinear grid, characterised by the presence of several urban voids and vacant land that affects the general permeability of the neighbourhood. The topography also determines an organic urban fabric, in which the size of the blocks and their occupancy rate are quite different. As for the quality of the streets, while a great part of the streets is paved and in good conditions, sidewalks are very narrow; moreover, several houses are not adequately connected to the main streets of the neighbourhood and several cul de sacs can be found.

Urban forms under the test of women's experiences: the ethnographic approach

The ethnographic analysis allows detecting more evident conditions and feelings of insecurity in mobility practices, intervening as restrictors of woman's mobility. Sensations of fear and insecurity can be traced back to three peculiar morpho-functional conditions affecting the three selected neighbourhoods, allowing to detect which peculiar condition plays a relevant role among the investigated 3D (the spatial layout of street networks, the land-use mix and street design and equipment).

In term of the *spatial layout of street networks*, the role of permeability emerges as a relevant issue from the women's mobilities experiences we collected (see Figure 5).

In Arborizadora Alta (Complex organic layout), the topographical conditions and the number of urban voids have a significant impact on determining low street permeability. This leads to low connectivity and isolation of large segments of streets so that women living within this neighbourhood have very few alternative routes for making their trips. Eventually, both women choose to cross the mountain, despite the discomfort caused by the daily activities, being thus forced to decide between comfort and safety. For instance, S. and M. (who live in the southern part of the neighbourhood) have few routes to access Transmilenio bus feeders when they go to work. To access the closest bus stop, women living in this part of the neighbourhood have two choices: crossing a steep mountain

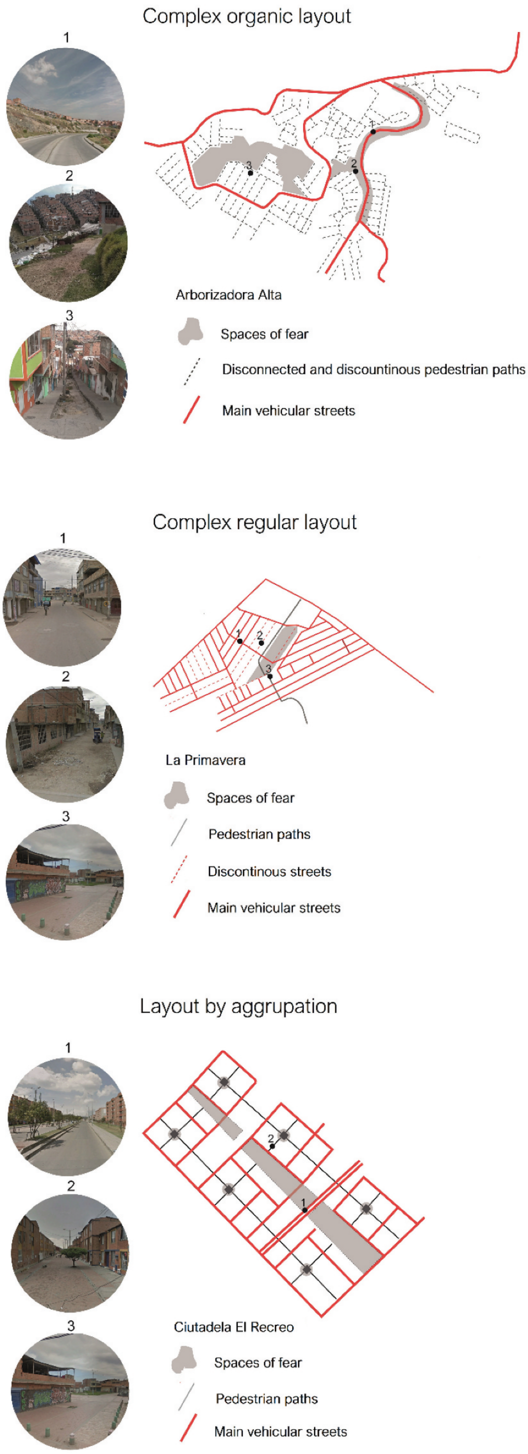

Figure 4. Design: street paths in the three selected neighbourhoods (our elaboration of data collected by the authors).

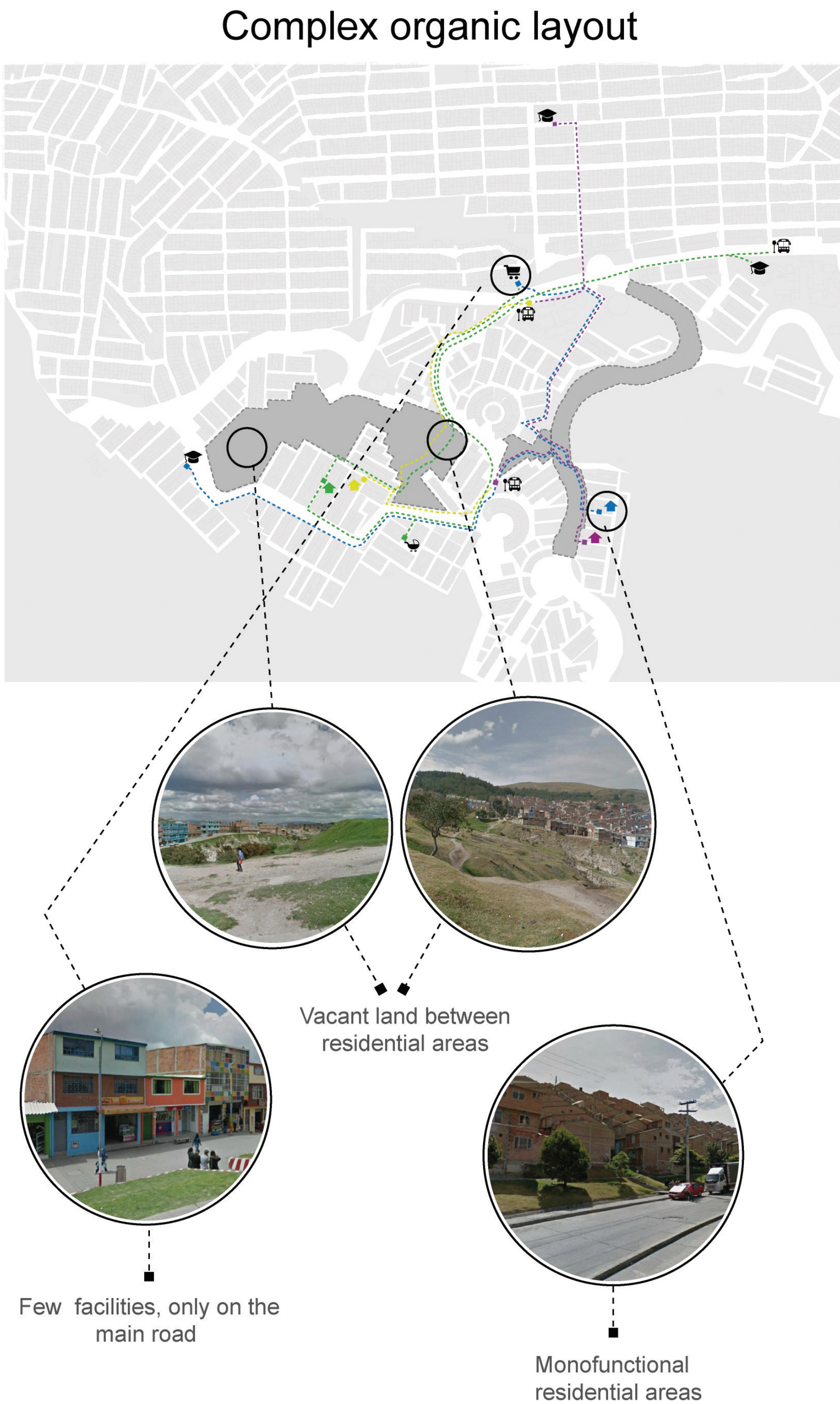

Figure 5a. Fear spaces related to built environment features mentioned by the interviewees (our elaboration).

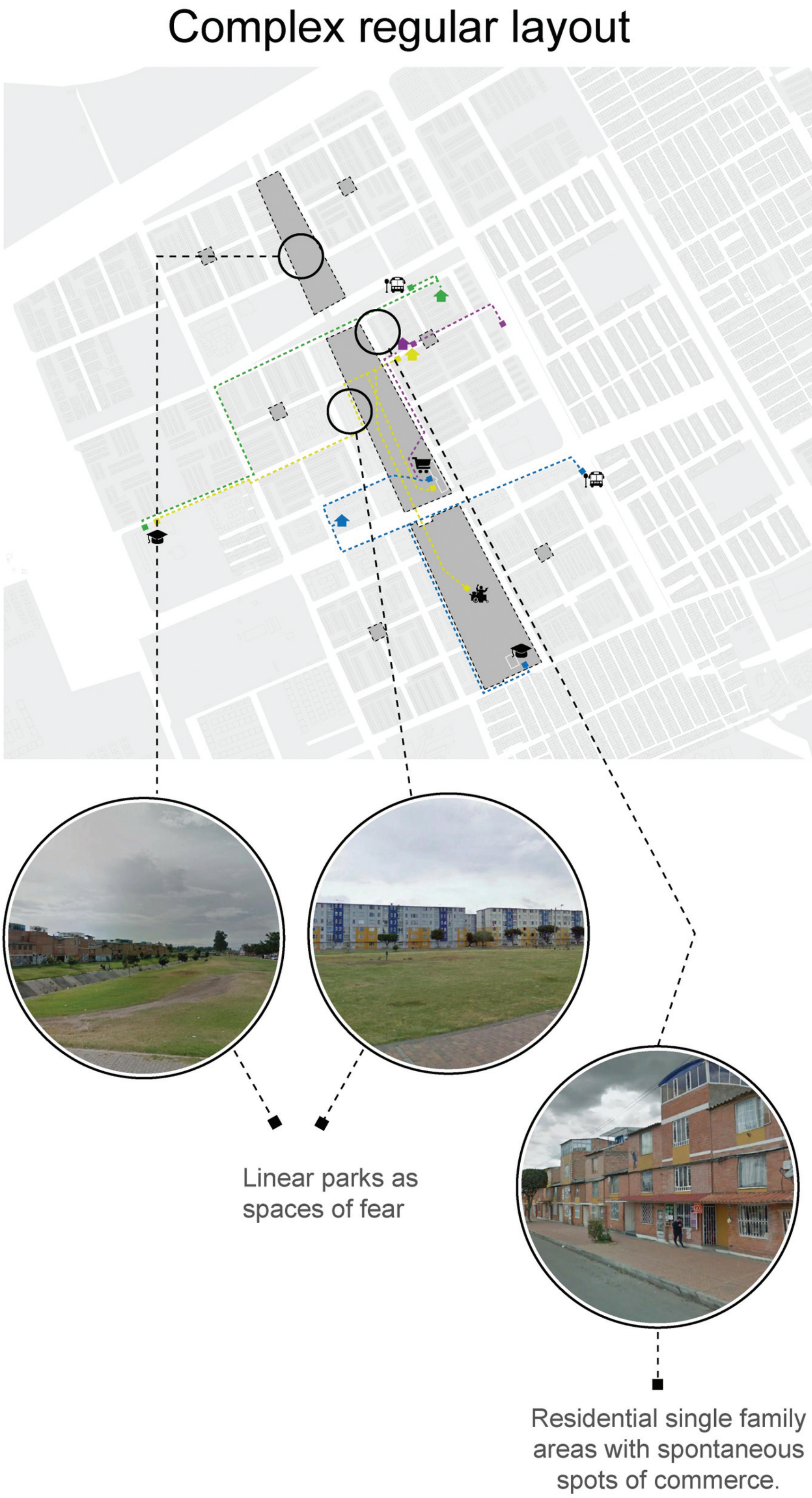

Figure 5b. (Continued)

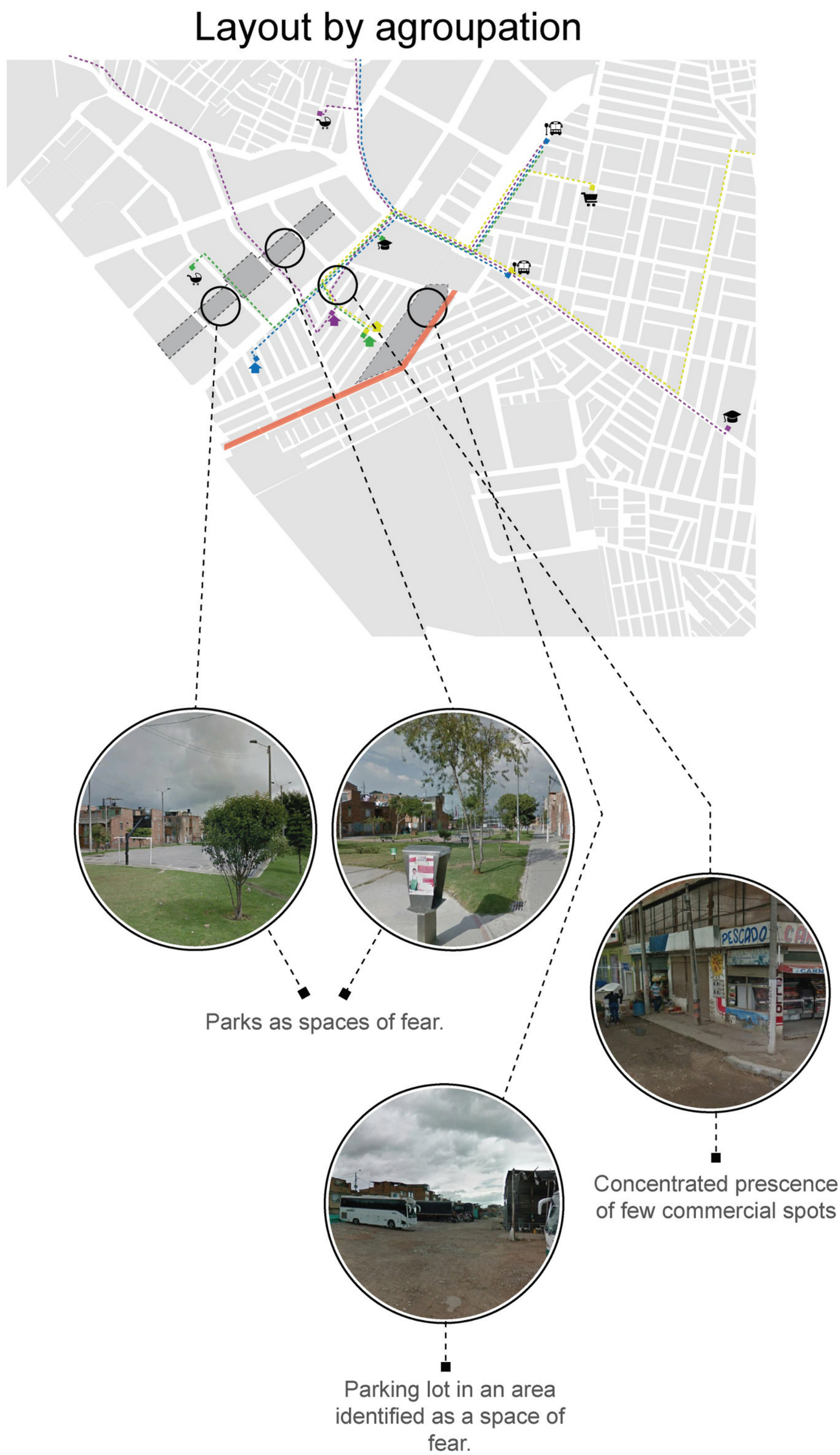

Figure 5c. (Continued)

described as highly uncomfortable or walking through the car road, described as lonely and risky due to the absence of any pedestrian flow (Figure 6) topography, because the route is shorter and relatively safer than the road option. As S. tells:

> crossing the mountain is exhausting because there is no clear path to walk. I could simply use the route of the road to access bus stops located in the main avenue of the neighbourhood, but the route is lonely, and it feels dangerous. Additionally, in the main avenue of the neighbourhood, some cheap shops and supermarkets are located, so, for accessing these facilities, I prefer to cross the mountain and find an alternative route and skip the route of the road which is more direct but very not safe. (S., 38 years old, single, school teacher)

On the other hand, the low equipment of the streets network, the lack of comfortable sidewalks and the problematic topography produce an additional burden on the realisation of care related activities: S. states that:

> the experience of walking through the neighbourhood is exhausting, especially as a mother of a six-year-old boy. I bought a stroller when I gave birth to my son, but I never use it. Is very difficult to carry a stroller within the neighbourhood due to the inclination of the streets and the width of the sidewalks, so generally many women simply have to carry their children in their arms everywhere, which is exhausting. (S., 38 years old, single, school teacher)

The previous situation described by S. reflects the low permeability of the urban form: the road network of the neighbourhood, with many cul-de-sacs converging in one street, with only two alternative options for reaching the bus stop: a tiresome walk crossing the mountain, or an insecure itinerary along the main street.

The organic urban system found in this neighbourhood typology also segregates residential areas, providing few alternative routes often characterised by rough paths, producing additional and eventual risks. As M. recalls:

> My mother-in-law, who is an older person, has had some physical accidents when walking through the route of the mountain, the slope is very steep and presents obstacles like stones. For this reason, when she has to buy groceries in the supermarket, she has to be accompanied. (M., 29 years old, married, domestic worker)

Instead, the neighbourhood La Primavera – characterised by small blocks – is physically much more permeable, but affected by symbolic barriers and spaces perceived as fearful. Consequently, to avoid such spaces, women are obliged to find alternative safer (but longer) routes. For example, this neighbourhood has an evident barrier, an empty lot that separates the neighbourhood from a stigmatised residential area, which female residents qualify as unsafe and violent. This area is identified as a space of fear since the residents associate it with criminal activity coming from across the frontier (Figure 6). Some neighbourhoods have witnessed armed robberies near the area identified as a space of fear, especially during early mornings, when residents have to leave their houses to go to work: as P. recalls:

> once I saw how some men in motorcycles attacked a girl at 6:00 am, a neighbour had to come out shoot a gun a few times for frightening the thieves. (P., 26 years old, married, graphic designer)

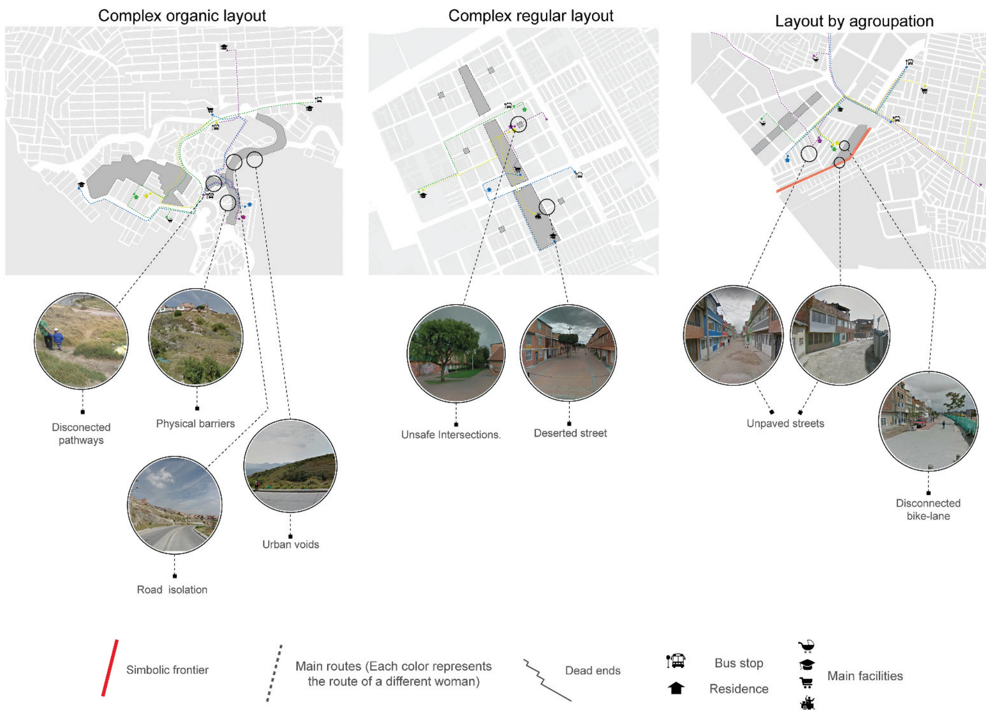

Figure 6. Fear spaces related to street layout features mentioned by the interviewees (our elaboration).

The existence of this frontier obligates female residents to walk long distances to avoid risks, but here women have more alternative routes, due to the high number of intersections and the local street layout: as a result, M. states that she prefers

> to go around the neighbourhood and find a less threatening route to reach certain places. (M., 31 years old, single mother, construction worker).

In Ciudadela El Recreo, the street layout configuration was designed around residential super-blocks to facilitate pedestrian mobility. Despite this initial intention, in the interviews is clear how women perceive this system of pedestrian streets as a potential place of risk, especially at night and early in the morning mornings. As G. states:

> every morning, I have to be accompanied by my mother while I walk toward the bus stop. I consider the internal pedestrian streets of the neighbourhood as dangerous. Specifically, the park located at the intersection of the alleys can be used by thieves to hide and attack you by surprise. (G., 27 years old, single, elementary teacher)

In this way, the conformation of the block, created to provide friendly pedestrians paths inside the super-block, ends up creating blind spots in the intersections of the streets, offering safe spots for possible crimes: as P. describes:

> *the space of the intersection located in the middle of the block is risky especially during the nights when visibility is very poor. The design of the corners of the block allows people to hide in these places*. (P., 28 years old, married, street vendor)

On the other hand, the neighbourhood is also composed of residential multi-family housing blocks, surrounded by fences that generate a different relationship with the block. The impenetrability of the latter through large segments of streets and the impossibility to create a more dynamic relationship between the block and the sidewalk increase the feelings of vulnerability, especially during the nights. In this case, the size of the blocks increases, and permeability decreases.

In term of land use mix, the neighbourhoods selected for this research can be considered as high-density areas, characterised by a robust commercial character and generally denser than the North American neighbourhoods. In la Primavera, despite the intense commercial activity found in the area, all the interviewees had a low perception of safety in the neighbourhood and the surrounding areas. In particular, the interviewees identified two types of areas to avoid while walking: recreational areas, such as parks and the bike lane; and an area with relatively low levels of activity, due to the location of a parking lot. As P. states:

> I avoid parks located near the neighbourhood because they are often frequented by drug users and thieves, especially during the nights. (P., 26 years old, married, graphic designer)

In Arborizadora Alta, the extensive presence of vacant green areas provokes the absence of activity in large segments of the local streets. M.F., who lives in a residential located next to a vacant green area, recalls that she has to be escorted by a family member when she arrives home at night:

> After arriving from work, I have to walk approximately 15 min to my residence, but at night the route is not very safe and lonely, so my father has to wait for me at the bus stop, and then he walks with me towards the house, so I can feel safe. (M.F., 23 years old, single, Factory worker)

In this case, vacant spaces creating frontiers, limit women's mobilities and become a source of fear.

Finally, in Ciudadela El Recreo a large number of green areas with undefined activity and the missing functional relationship between public spaces and residential areas create barriers for safe and fluid movement. In general, local facilities are not close to the spaces identified as unsafe. As G. mentions:

> Despite the existence of the linear park, I prefer to make recreational activities outside the neighbourhood. Generally, when I walk at nights, I prefer to avoid the routes located next to the linear park because some parts remain deserted and spaces are not sufficiently illuminated. (G., 27 years old, single, elementary teacher)

On the contrary, A. uses the park during the weekends to walk with her daughter, she feels that

the public spaces located in the neighbourhood are beautiful, but I avoid walking through them at night. (A., 19 years old, single, accounting student). According to the interviews, specific land uses can have ambiguous effects: for example, empty parks can attract some users while reducing the propensity of others to walk through them. This situation appears in three case studies. During the interviews, most women declared not to use local parks, especially in the case of Arborizadora Alta and La Primavera. There are two main reasons: first, some of the parks were not adequate nor properly equipped for specific uses, such as children's play; second, other users, such as drug addicts or thieves waiting to assault passers-by, may monopolise the parks.

Regarding *the street design and equipment*, the visibility and the lighting provision (Figure 7) play a relevant role considering also that in Bogotá, daily mobility becomes more complicated during the dark moments of the day (from 6 pm to 6 am). Especially in Arborizadora Alta, these conditions reduce women's mobilities at night. In some cases, women need to be accompanied by other members of the family, commonly a man:

> After coming back from work, my dad has to pick me up at the bus station when is possible, because the route home is lonely and very bad illuminated (M.F., 23 years old, single, factory worker).

> During the night, after coming back from work, I have to cross the mountain again. On these occasions, this route becomes dangerous due to the lack of adequate lighting. I have to cross the mountain as fast as I can because I feel scared. (S., 38 years old, single, school teacher).

In each interview, women admitted feeling higher levels of anxiety, when walking from the transit stop to home during the late nights, especially after 9 pm, when all the local commercial activities are closed and the feeling of emptiness increases.

In the three selected neighbourhoods, safety perception emerges thus as a mediator between built environment attributes and walking behaviour, guiding the choices of the paths in the utilitarian and non-utilitarian trips. For utilitarian trips, the avoidance of specific routes, due to the presence of fear spaces, causes ripple effects on the time spent in a displacement or the access to opportunities in the city. The presence of children, who rely on women for moving – especially early in the morning, increases the perception of insecurity; moreover, night hours are considered as the most dangerous time of the day. In the non-utilitarian trips, women avoid certain places, such as recreational areas. Several factors determine such avoidance: the absence of activities during the day, the absence of lighting or the presence of barriers that limit the routes, and the deterioration or the monopolisation of the space by certain undesirable users. The interviewees take travel decisions balancing the optimal proximity to a place, physical

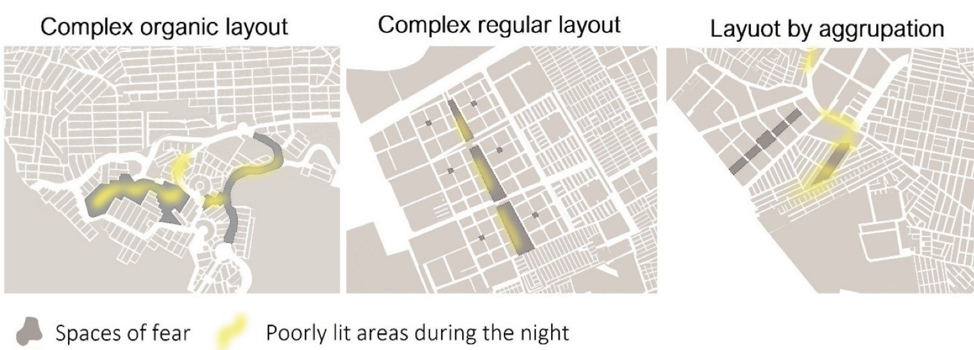

Figure 7. Fear spaces and areas with insufficient lighting mentioned by the interviewees (our elaboration).

comfort and personal safety, being the latter the most important reason for travel choice. Instead, the most significant triggers of fear perception are isolation, lack of pedestrian flow or activity and poor visibility of places. And, despite referring to different kinds of the built environment, the mobility experiences of the interviewed women share a prevalence of negative emotions (see Figure 8 for a synthesis). The interviewees mention how their routes often trigger feelings of anxiety and fear, leading to low levels of perceived safety, attributable to three aspects closely related to the morpho-functional patterns: the spatial layout of street networks, the land-use mix and street design and equipment.

Figure 8a. Synthesis of the fear spaces mentioned by the interviewees (our elaboration).

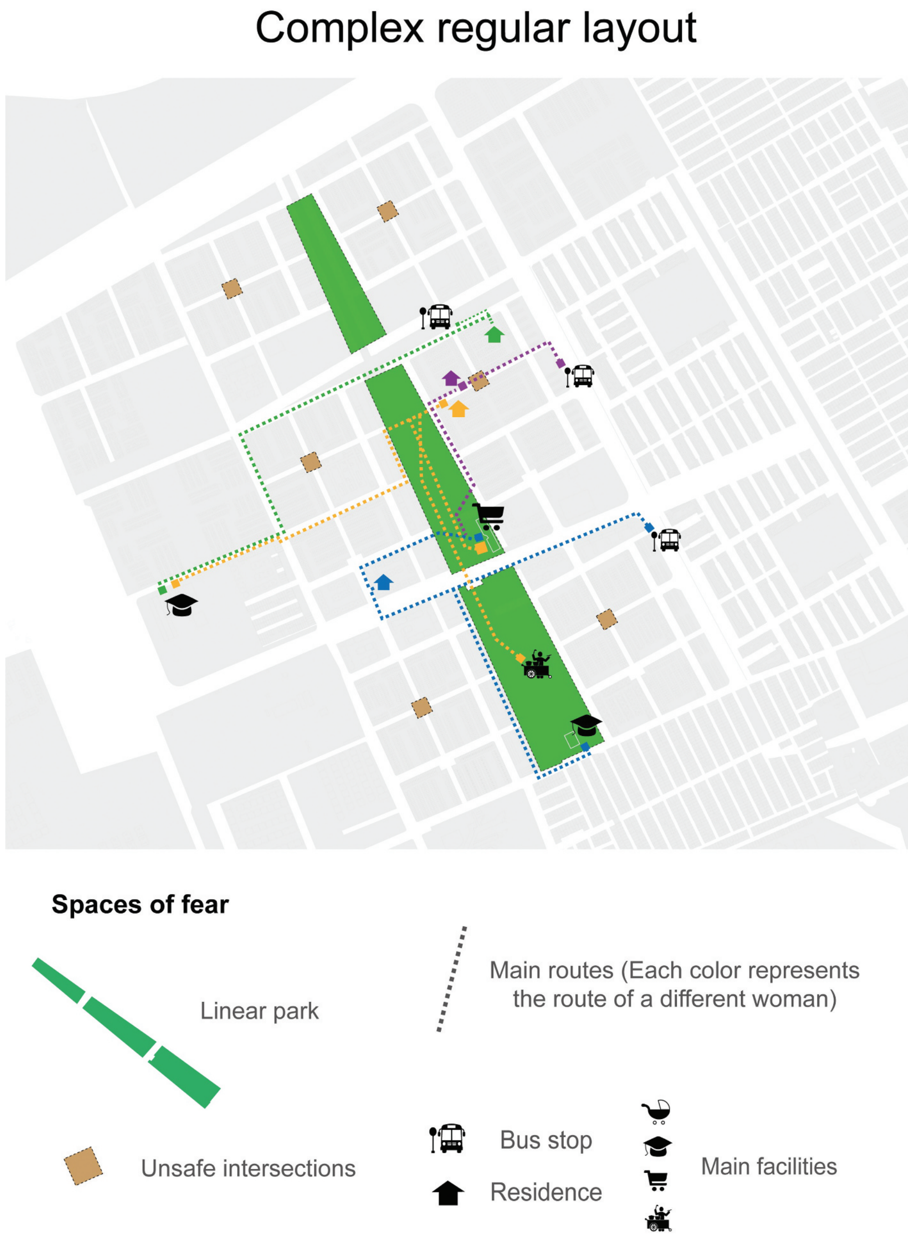

Figure 8b. (Continued)

Figure 8c. (Continued)

Conclusions: women and the production of urban space through mobility

Reading the urban forms through the mobility experiences of women shows that – also in informal contexts such as those investigated – standardised urban design understates women's needs and behaviour, contributing to reproduce constant spatio-temporal barriers that constrain the liveable and safe women's mobility. This condition has more onerous consequences, especially in the contexts such as those investigated in Bogotá, where women's mobilities become even more difficult because of the scarce provision of public transport, the lack of urban equipment and the unsafe context conditions. Combining the morpho-functional features of the three neighbourhoods with the women's perceptions and needs, we can relate the women's urban experiences to specific features of the urban form, and highlight, relevant elements for addressing gender-oriented urban policies.

Among them, the urban layout permeability plays a significant role in terms of wayfinding opportunities. Topographical barriers, isolation of large street segments, as well as many *cul de sacs* define a less integrated street network in which fewer route choices are guaranteed. To address these issues it is necessary, for example, to improve connections of main roads with unfinished pedestrian paths, ensuring pedestrian continuity among different residential areas, the public spaces and the public transport stops. An urban fabric composed of short blocks avoids isolation of large segments of streets, while higher chances to turn corners maximise the possibilities for women to reduce risks and escape from danger.

The design and the maintenance level of streets and especially sidewalks is a relevant element for accommodating specific needs related, for example, to maternity. As visible in the diaries description, currently, for women it is challenging to travel throughout the neighbourhood with a stroller, due to the width of the sidewalks, the presence of obstacles like garbage cans and the lack of ramps to access to residential areas, conditions that reduce even more the choice of alternative routes. Other relevant dimensions are the improvement of the lighting conditions, ensuring a homogenous distribution of lighting through streets, corners and parks; street furniture and design equipment not acting as obstacles; the maintenance of public spaces in term of adequate cleaning and care for avoiding a permanent state of abandonment and lack of appropriation. These are rather obvious conditions but are crucial to prevent the spreading of illegal activities.

Alongside the layout and the equipment of the paths network, an important role is played by the land use mix. A balanced land use mix allows to have more opportunities available at the local scale while having also a positive effect on natural surveillance and safety perception (Christian et al. 2011): these are elements that contribute to the walkability of a place. However, our case study partially questions this shared assumption: despite the high commercial-retail diversity (above all in the case of La Primavera), women still found barriers to walking and were prone to feel unsafe, especially at night and early in the morning. Measures of land use mix – like the entropy index – can be misleading, since the existence and proportion of specific land use in an essential but insufficient condition: other crucial features are their distribution, the opening time of the activities, and the interaction with the public spaces and their features.

Moreover, the results show that in the examined settings women consider public spaces as fear spaces, questioning the usually positive value that is associated to them. The presence of many small parks within the area in fact creates several 'blind spots' that remain empty during the day, due to the lack of equipment and adequate activity attractors during the week, especially during the nights. To counterarrest this negative perception of public spaces, it is relevant to consider that the size and the location of parks within a determinate area, as well as the activities within the open areas and their functional relationship with adjacent blocks (Groff and McCord 2012), are relevant site-based conditions in the design of safe and liveable spaces for women, becoming crucial to avoid the formation "junkspaces" (Koolhas, 2006). Similarly, land use mix provides natural surveillance only during a few hours of the day. The neighbourhoods remain inactive at night and early in the morning when some of the interviewees move. The presence of facilities with alternative night use, such as schools used at night for adult education programs or lifelong education initiatives, can provide natural surveillance in public spaces.

Alongside conditions aimed at supporting women's walkability in a friendly environment, also gender-tailored mobility services represent a basic issue and require the definition of mobility measures able to accommodate the various activities of which women are in charge. In different contexts around the world, examples of such improvements are multipurpose compartments, with additional space reserved for strollers or wheelchairs; low-floor buses allowing easy boarding and exiting; barrier-free, welcoming and pleasant stations and stops with clearly visible transport information; and routes with the possibility of requesting stops.[4] While these measures have a positive impact on female transit riders (Loukaitou-Sideris 2012), in Bogotá the improvement of the basic transport conditions in peripheral areas, as in the three investigated neighbourhoods, and especially during the night, becomes a necessary policy to avoid the women's immobilities. Two interventions may contribute to widen access to urban opportunities and alleviate woman safety problems: expanding the Transmilenio bus feeder network, which currently does not cover enough parts of the examined peripheral areas, and implementing a request stop program – allowing women to disembark at any point of the route which is closer to their final destination. These measures could facilitate the compulsory mobility of women, who have to move at night when there are no eyes on the street are not possible, and the streets lack vitality.

Examining mobility from the perspective of women in a specific geographical context as the peripheral areas of Bogotá, three challenges emerge, which can be relevant also for future studies. First, the adopted approach challenges consolidated analytical and design tools, which are based on a man-centred perspective, suggesting to experiment and adopt methodological approaches able to interpret the women's territorialities,[5] as well as their needs and opportunities, especially in informal and peripheral neighbourhoods. Second, the approach allows giving visibility to women as producers of urban space, assuming that women do not only move their body in the space but also mobilise resources and capitals while, at the same time, producing spaces (Piscitelli 2018). Third, the approach highlights the mobile female urbanity as a heuristic tool for defining more effective urban and mobility policies, based not only on the problems and needs of women but also on their tactics of adaptation to the constraints and difficulties, allowing also the experimental implementation of tactical urban interventions.

We are aware that the approach followed has some limitations. First of all, the sample size of the interviewees is very limited, despite its selection based on some relevant criteria affecting woman's mobility (i.e. working women with children, both single and married, involved in the fulfilment of household needs, with an age from 18 years old to 50 years old, with regular work-related mobility patterns). This means that the proposed approach may not be extensively used for depicting current forms of inequality in women's mobilities, but it offers some insights into the design of safe and liveable public spaces in informal settlements, where some rules applicable to planned contexts risk producing unexpected and negative results. As is well known, ethnographic analyses with travel along and diaries represent an extremely powerful analytical tool, despite being expensive and difficult to apply to large samples where even the use of digital surveys is not feasible. In our research, they provide micro-stories, profiling different woman's mobility practices in deprived districts (Vecchio 2020). Moreover, it could be relevant to expand the analysis assuming an intersectional perspective, able to go beyond low-income women: in this sense, it could be significant to consider how other variables (such as age or ethnicity, or even a more complex, non-binary understanding of gender, see for example Ritterbusch 2016) affects mobilities and safety perception. Finally, it must be noted that the three examined settings may not be paradigmatic situations of informal settlements in the Global South, even if they provide a significant example of what forms of mobility and safety acquire in deprived neighbourhoods with low access to transport. Thus, our work is one more small contribution to the necessary understanding of how the built environment influences women's mobilities, making it necessary to develop more advanced research on the issue and experiment with gender-sensitive approaches to urban and mobility issues.

Notes

1. The municipal land use plan of Bogotá – the POT, Plan de Ordenamiento Territorial – identifies seven morphological units, based on indicators like historical growth of the city, the socioeconomic strata, the average size of blocks, the number of lots per block, average occupation per block and the topography.
2. According to Marshall and Çalişkan (2011), urban morphology is the "science of urban form and structure" that includes analysis, description, possibilities, and explanation of generation processes.
3. Among them: IDECA- Infraestructura de Datos Espaciales para el Distrito Capital/ Spatial Data Infrastructure for the Capital District; Encuesta multipropósito 2017 / multipurpose survey 2017; Secretaria distrital de Planeación/ District Planning Secretariat; DANE- Departamento administrativo nacional de estadística/ National Administrative Department for Statistics.
4. This measure has been adopted in Toronto and later implemented by several transportation authorities (for example in London and New York).
5. For the concept of territoriality see Raffestin (2012).

Highlights

- We investigate the effects of the built environment and urban form on female mobility
- We combine a morpho-functional analysis with an ethnographic survey
- We identify potentials and barriers in the urban form from a gender perspective
- Perceived safety acts as a mediator between built environment attributes and female walking behaviours
- We propose policy orientations based on female mobilities

Disclosure statement

No potential conflict of interest was reported by the author(s).

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